

TECHNOLOGICAL DESIGN

Technological design courses provide students with a variety of learning experiences that focus on the practical application of the principles of engineering, architecture, and design. These activity-based courses emphasize problem solving to meet design challenges in a wide range of areas, which may include apparel and textile design, architectural design, interior design, mechanical and industrial design, and robotics and control systems.

Students learn to apply knowledge of research, historical trends, design, materials, fabrication methods, and testing criteria to develop innovative and environmentally sustainable products, processes, and/or services. The technologies and processes used to create design solutions may include both traditional and computer-based drafting methods, scale models, working prototypes, animations and simulations, displays, portfolios, and presentations.

The list of approved emphasis areas for technological design can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Technological Design and the Environment, Grade 11

Open

TDJ30

This course enables students to apply a systematic process for researching, designing, building, and assessing solutions to address specific human and environmental challenges. Through their work on various projects, students will explore broad themes that may include aspects of industrial design, mechanical design, architectural design, control system design, and/or apparel design. Students will develop an awareness of environmental and societal issues related to technological design, and will learn about secondary and postsecondary pathways leading to careers in the field.

Prerequisite: None

A. TECHNOLOGICAL DESIGN FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** describe the design process, and identify ways in which technological design can address an environmental need or challenge;
- A2.** describe and apply strategies, techniques, and tools for researching, planning, and organizing projects to meet a specific environmental or other need;
- A3.** demonstrate an understanding of drafting standards, conventions, and guidelines for representing design ideas graphically;
- A4.** compare various kinds of models and prototypes and identify criteria, equipment, and methods for assessing them;
- A5.** demonstrate an understanding of the technical and environmental terminology and the communication methods and formats used in the design process.

SPECIFIC EXPECTATIONS

A1. Design Process

By the end of this course, students will:

- A1.1** describe the purpose of design for a given project (*e.g., cleaner energy, reduced carbon footprint, less manufacturing waste*) in terms of key technological concepts (*e.g., aesthetics, control, environmental sustainability/stewardship, ergonomics, fabrication, function, innovation, material, mechanism, power and energy, structure, safety, systems*) (see pp. 7–8);
- A1.2** describe the need for technological designs that take environmental factors into account (*e.g., fuel-efficient vehicles, non-toxic paints and pesticides, renewable energy sources, sustainable production of materials*);
- A1.3** identify and describe the steps in the design process (*e.g., define the problem or challenge, taking into account relevant contextual or background information; gather information about criteria, constraints, and available materials; generate possible solutions, using techniques such as brainstorming; choose the best solution; develop and produce a model or prototype; test the model or prototype; incorporate improvements or redesign and retest; report results*) (see pp. 22–23);
- A1.4** describe the relationship between various steps of the design process (*e.g., testing a model*

or prototype, and then incorporating improvements or redesigning and retesting if necessary).

A2. Research, Planning, and Organization

By the end of this course, students will

- A2.1** identify and apply strategies for gathering information from various sources (*e.g., books, Ministry of the Environment website, interview with a naturalist*) for a design project that meets an environmental need;
- A2.2** describe strategies and tools for planning and organizing a design project;
- A2.3** plan ways to apply the principles of sustainability and minimize environmental harm throughout the design process for a project (*e.g., plan to use recycled materials, limit the use of energy-consuming equipment*).

A3. Representing Design Ideas Graphically

By the end of this course, students will:

- A3.1** compare different methods for representing design ideas graphically (*e.g., mind maps, sketches, computer-aided drawings*);

A3.2 identify and describe various types of technical drawings (*e.g., 2D and 3D drawings, floor plans, elevations, sections, detail and rendered drawings*) and the drafting standards and conventions used in them (*e.g., symbols, abbreviations, line types*).

A4. Making and Testing Models and Prototypes

By the end of this course, students will:

A4.1 compare various kinds of models (*e.g., conceptual, physical, virtual, theoretical*) and prototypes (*e.g., proof-of-principle, functional, form study*), taking into account the tools and materials required to produce them (*e.g., tools: cut out patterns, animation; materials: paper, wax, clay, wood, metals, composites, plastics, cardboard, starch, textiles*), as well as the amount of energy used and the wastes produced;

A4.2 identify criteria for assessing designs (*e.g., adherence to specifications, effectiveness, cost, durability, appearance*);

A4.3 identify criteria for assessing the environmental friendliness of a design and of the processes required to produce it (*e.g., by-products, waste, energy consumption, reuse and/or recycling of materials, biodegradability*);

A4.4 identify equipment (*e.g., decibel meter, height gauge, weighing scale, colour wheel*) and methods (*e.g., measuring dimensions, compression testing, wind tunnel testing*) used for assessing prototypes.

A5. Reporting and Presenting

By the end of this course, students will:

A5.1 demonstrate an understanding of technical and environmental terminology used in the design process (*e.g., drafting versus drawing, scale versus ruler, greenhouse gases, parts per billion, fossil fuel*);

A5.2 describe how various report formats can be used to record design ideas and document the progress of a design project (*e.g., portfolio, journal, logbook, sketchbook*);

A5.3 choose the most effective method to communicate progress and results for a given project from a variety of presentation tools and strategies (*e.g., LCD projector, display board, oral or written presentation*).

B. TECHNOLOGICAL DESIGN SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** use appropriate tools and strategies to research, plan, and organize design projects that have environmentally sound designs and production processes;
- B2.** apply appropriate methods for generating and graphically representing design ideas and solutions;
- B3.** construct models and prototypes using a variety of techniques, tools, and materials, and assess these models and prototypes in terms of the design criteria;
- B4.** report on the progress, environmental rationale, and results of the design process, using appropriate formats and styles.

SPECIFIC EXPECTATIONS

B1. Research, Planning, and Organization

By the end of this course, students will:

- B1.1** gather and summarize relevant information for developing various designs (*e.g., Canadian Standards Association [CSA] publications, Ontario Building Code, environmental criteria*);
- B1.2** investigate and describe economic and environmental factors that should be considered during the design process;
- B1.3** select and apply effective planning and organizational tools and strategies (*e.g., sequence chart, time sheet, swatch book, checklists, file management*) to develop environmentally sound design projects.

B2. Developing and Representing Design Ideas

By the end of this course, students will:

- B2.1** create freehand sketches showing key features (*e.g., shapes, texture, protrusions, unusual features, relationships among parts*) to help develop initial concepts for a design project;
- B2.2** apply mathematical and scientific concepts and skills as required in the course of designing various products and/or processes;

- B2.3** produce hand-drafted and/or computer-based technical drawings (*e.g., elevation, plan, schematic, exploded view*) of design solutions, using common drafting standards and conventions (*e.g., format, line type and weight, symbols*).

B3. Making and Testing Models and Prototypes

By the end of this course, students will:

- B3.1** use appropriate tools, equipment, and materials to create design models and prototypes (*e.g., tools: rotary tool, bandsaw, fabric shears, computer-aided design [CAD] software; equipment: computer, sewing machine; materials: paper, wax, clay, wood, metals, composites, plastics, cardboard, starch, textiles*);
- B3.2** use appropriate metric and imperial measuring tools, scales (*e.g., metric: 1:10, 1 cm:1 m or 1:100, 1:500; imperial: 1/2":1' or 1:24*), and proportion techniques when creating a model or prototype;
- B3.3** test models and/or prototypes, and evaluate designs using student-generated criteria (*e.g., by-products, waste, energy consumption, biodegradability, reliability, durability*).

B4. Reporting and Presenting

By the end of this course, students will:

B4.1 present a report summarizing the design choices, progress, and results of the design project, with an emphasis on how the design deals with environmental concerns, using a variety of tools (*e.g., presentation software, interactive white board, web pages, word-processing software*);

B4.2 report and reflect on the decisions they made and their experiences throughout the design process, using appropriate written and/or oral formats.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of environmentally responsible design practices and strategies, and apply them in the technological design process and related activities;
- C2.** describe how society influences technological innovation and how technology affects society.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** identify environmental issues that affect technological design (*e.g., pollution, greenhouse gases, resource use, ozone depletion*);
- C1.2** describe and apply best practices for conserving energy and other resources when designing a product or process (*e.g., reuse or recycle materials, use wood glue instead of hot glue, use energy management systems for computers*);
- C1.3** research and report on organizations and/or community partners that foster environmentally friendly design practices (*e.g., Ontario Centre for Environmental Technology Advancement, other Canadian environmental technology advancement centres*);
- C1.4** describe innovative technological designs (*e.g., alternative energy sources, more efficient automobiles, ways of reducing manufacturing waste*) that were developed in response to changes in the environment (*e.g., global warming, pollution, rainforest destruction*);
- C1.5** compare the environmental impact of various products that are used for the same purpose (*e.g., plastic bags versus paper bags or reusable cloth bags, paper cups versus polystyrene foam cups*).

C2. Technology and Society

By the end of this course, students will:

- C2.1** research and report on how society influences technology (*e.g., higher energy costs spur development of more efficient vehicles, increasing population density leads to the construction of taller buildings, environmental awareness leads to development of alternative energy sources*);
- C2.2** describe how technological design has benefited society (*e.g., prosthetic limbs, on-demand water heaters, catalytic converters, nanotechnology, wireless communication, air purifiers*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** describe and apply appropriate health, safety, and environmental practices and standards throughout the design process;
- D2.** identify careers related to technological design, and describe the training and education required for these careers.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** investigate and describe health, safety, and environmental laws, regulations, standards, and agencies that can affect technological design (e.g., *Ontario Environmental Bill of Rights, Clean Water Act, Canadian Standards Association [CSA] standards, Workplace Hazardous Materials Information System [WHMIS]*);
- D1.2** demonstrate an understanding of and follow personal and environmental health and safety procedures with respect to processes, materials, tools, equipment, and facilities throughout the design process and when performing related activities (e.g., *use protective equipment; set tool and equipment guards properly; ensure adequate ventilation and ergonomic seating and other workplace arrangements; follow safe operating procedures; keep work areas clean and organized; store materials and dispose of wastes properly*);
- D1.3** use protective clothing, gear, and equipment appropriately (e.g., *dust mask, safety glasses*).

D2. Career Opportunities

By the end of this course, students will:

- D2.1** identify a variety of career opportunities related to technological design (e.g., *civil engineer, architect, mechanical engineering technician, environmental technologist, landscape designer, fashion designer, interior designer*);
- D2.2** identify education and training requirements (e.g., *degree, diploma, certificate, apprenticeship*) for specific careers related to technological design;

D2.3 identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the technological design industry (e.g., *mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);

D2.4 demonstrate an understanding of and apply the Essential Skills that are important for success in the technological design industry, as identified in the Ontario Skills Passport (e.g., *reading text, writing, document use, measurement and calculation*);

D2.5 demonstrate an understanding of and apply the work habits that are important for success in the technological design industry, as identified in the Ontario Skills Passport (e.g., *working safely, teamwork, reliability, initiative, customer service, entrepreneurship*);

D2.6 maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in technological design (e.g., *work logs, skills checklist, sketches, drawings, photographs of models and prototypes, virtual models*), and explain why having a current portfolio is important for career development and advancement.